

AI and Robotics

Lesson 1a: Introduction

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Introduction

A little bit about me

- I am a member of the Taub Computer Science Faculty at the Technion.
- I lead the Collaborative AI and Robotics (CLAIR) lab
- My research focuses on multi-agent AI and multi-robot systems.
- My work offers novel approaches to facilitating collabroations between AI agents via the **design** of the environments in which they are intended to act.
- My email: `sarahk@cs.technion.ac.il`

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Acknowledgements

This course is inspired by two courses:

- Autonomous Robot Systems (CS189) I took at Harvard by Radhika Nagpal (now at Princeton)

  `https://www.radhikanagpal.org/`
- Multi Robot Systems by Amanda Prorok from Cambridge University

  `https://www.youtube.com/playlist?list=PLaTKfS3-bDpDyOwrxLcQRGxY9XJw33ANo`
- Cognitive Robotics by Erez Karpas from the Technion

  `https://www.youtube.com/playlist?list=PLaTKfS3-bDpDyOwrxLcQRGxY9XJw33ANo`

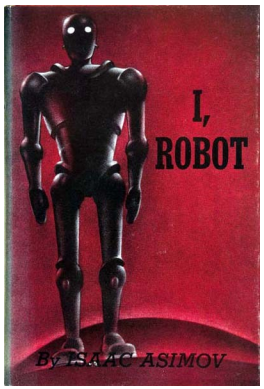
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History of Robotics

What is a robot ?

The word 'Robot' was coined by a Czech playwright named Karel Capek in his 1921 play "Rossum's Universal Robots"
Rabota = menial work
The word "Robotics" believed to be coined by Isaac Asimov





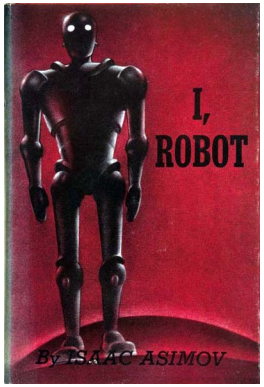
Definitions?

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Definitions?

- **Definition:** A **robot** is an autonomous system that exists in the **physical** world, can sense its environment, and can act on it to achieve some goals.
- **Definition:** An **autonomous robot** acts based upon its decisions, and is not controlled by a human.

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A Brief History of Robotics

Robotics emerged from the influences of:

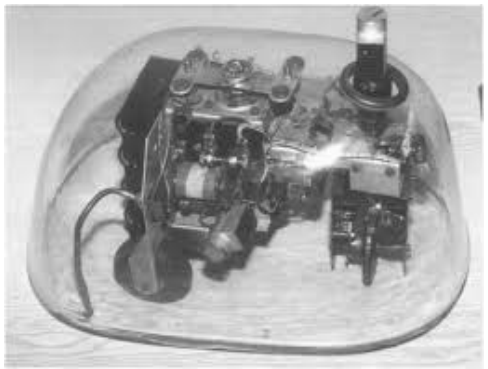
- **Control Theory:** the mathematics of controlling machines.
  - Develop algorithms governing to drive the system to a desired state, while minimizing delay, maximizing stability, etc.
  - In robotics,feedback control used sensors and making adjustments to keep the measured variable within a set range
- **Cybernetics:** the study of control and communication in biological systems and machines
  - Integration of sensing, action, and the environment
- **Artificial Intelligence:** the mechanisms for planning and reasoning;
  - The study of intelligent agents, which refers to any system that perceives its environment and takes actions that maximize its chance of achieving its goals
  - The original focus was on creating machines that think like humans (i.e., that pass the Turing test) - today the focus has changed.

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The First Robot

- Grey Walter (1910-1977) - a neurophysiologist and cyberneticist who created in the 1940s simple robots based on reflexes (before AI existed) - bio-mimetic machines.
- The **Tortoise** robot:
  - tricycle-like design (one motor per wheel)
  - one photocell: one bump sensor
  - rechargeable battery
  - analog circuit and vacuum tubes connecting sensors with wheels
- Behaviors:
  - find light / head towards light / back away from light
  - turn and push to avoid obsacles recharge battery
- Control: reactive, with prioritized reflexes
- First example of 'artificial life': Tortoises now shown in museums around the world





https://youtu.be/lLULRlmXkKo

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Artificial Intelligence

- Originate in 1956 at a workshop at Dartmouth University, USA: **Dartmouth Summer Research Project on Artificial Intelligence**
- Some of the researchers present: Marvin Minsky, John McCarthy, Allan Newell, Herbert Simon, Claude Shannon, John Nash, etc..



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Artificial Intelligence

- Outcome of meeting - what we need for AI is:
  - Internal models of the world
  - Search through possible solutions
  - Planning and reasoning to solve problems
  - Symbolic representations of information
  - Hierarchical system organization
  - Sequential program execution

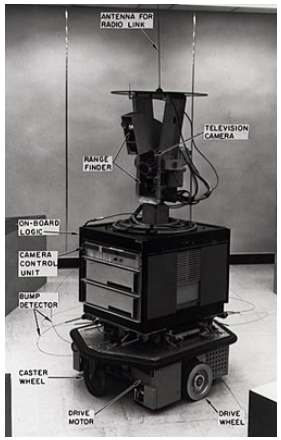


” We propose that a 2-month, 10-man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College. ”

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Shakey: the First intelligent robot

- Built in the 1960s at the Stanford Research Institute (SRI)
- The robot had
  - Camera and contact sensors
  - Application of AI techniques to vision algorithms
  - Lived in a simple black and white world
  - The robot shook when trying to execute its navigation plans.
- Reasoning:
  - Planning with STRIPS
  - Programming with LISP

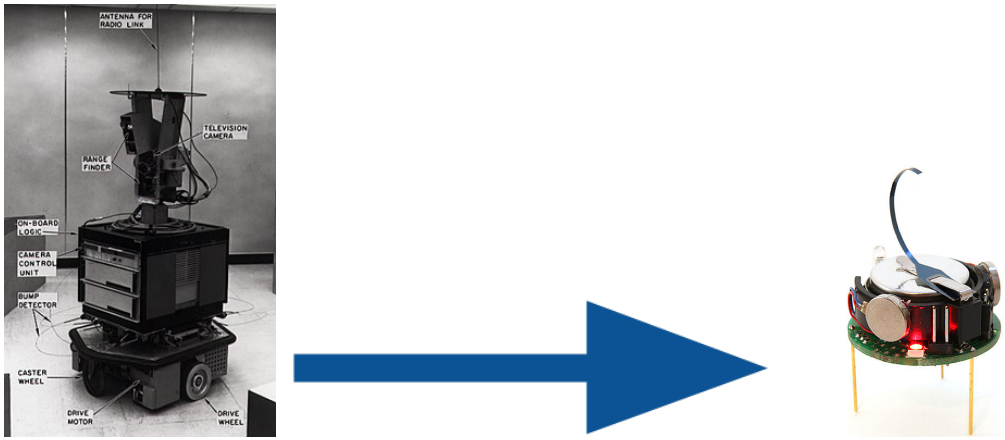


https://www.youtube.com/watch?v=7bsEN8mwUB8&t=21s

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A New Paradigm: Behavior Based Robotics

- Robots developed based on classical AI techniques in the 1970s and 1980s thought to be **too hard** and **too slow**.
- 1980s robotics enters a new phase:
  - reactive control
  - hybrid control
  - behavior-based control
- In stark contrast to purely **deliberative** control



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Behavior Based Robotics

A new wave in AI in the 80s-90s

- Spearheaded by Rodney Brooks (MIT) and later Ronald Akin (Georgia Tech)
- Stepping away from complex AI, deliberative planning (e.g., a chess playing computer) to the basics of intelligence.
- How? By realizing that organisms in nature follow simple sets of **reactive rules** giving rise to **reactive behaviors**

Which behavior-based robot do we all know ?

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Behavior Based Robotics

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Which behavior-based robot do we all know ?



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Behavior Based Robotics

Behavior-based architectures had a huge impact on collective intelligence and swarm robotics

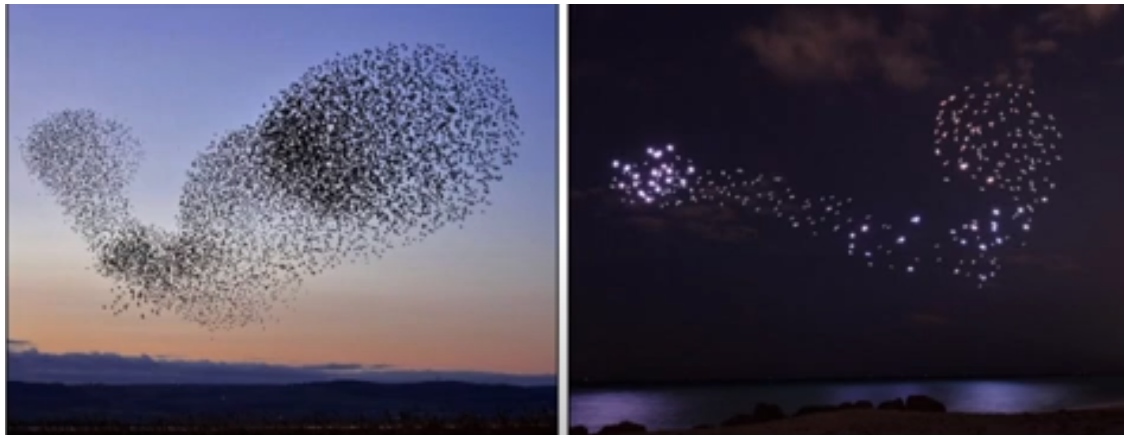


Figure 1: Starlings

`https://youtu.be/0bRocfcPhHU?t=255`

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Cognitive robotics

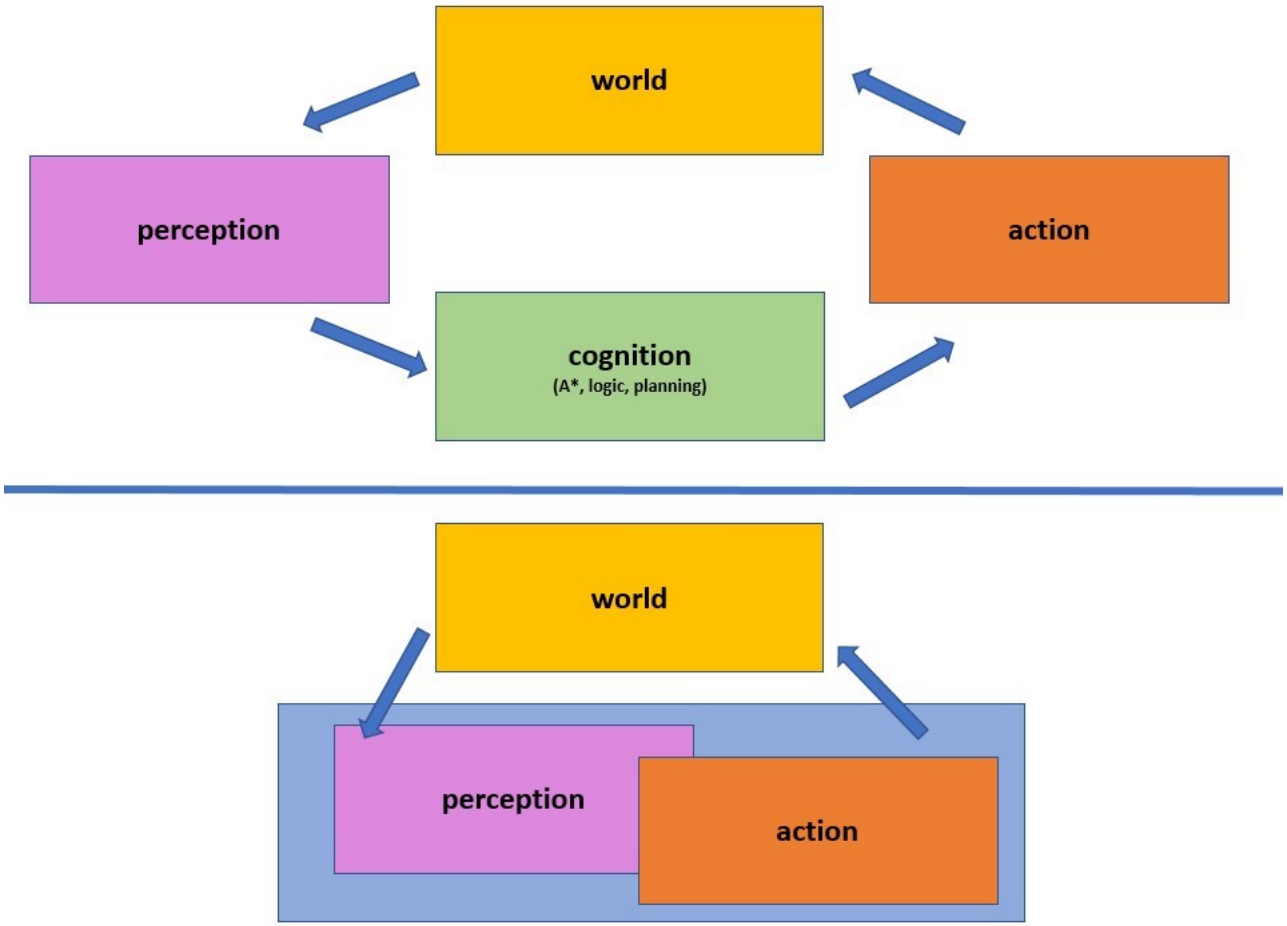
- A subfield of robotics concerned providing a robot with a processing architecture that will allow it to learn and reason about how to behave in response to complex goals in a complex world.
- Needs to
  - monitor itself for potential problems
  - schedule tasks in time
  - come up with novel plans to achieve desired goals over time
  - collaborate with other (autonomous) agents
  - deal with risk ...





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Two streams of thought in robotics



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Historical Overview

1990s to Today - A Few Highlights

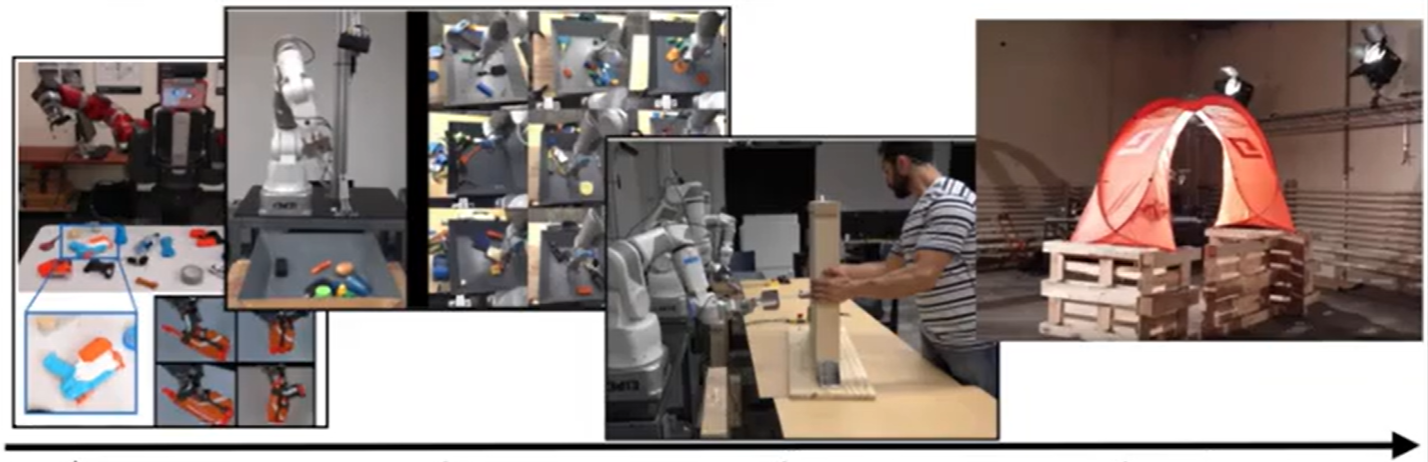


2015 and onwards: trend away from first principles-based approaches to data-driven approaches (black box control)

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Historical Overview

Some recent developments…



2015: Baxter takes 10 days using DL to learn to grasp

2016: Visual MPC with deep predictive models

2017: Learning based on human guidance

2018: drone racing

2015 and onwards: leveraging ML to learn perception as well as end-to-end control.

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Even More Recent



Figure 2: Today's robots

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Physical AI



https://youtu.be/cRZNwgvcWUg?si=kIfNi9zTUdHsXraJ&t=787

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Example



”Pick up an apple”



”Pick up an apple”

”I am hungry”



Example

Example

The person: ”pick up an apple”



What do we need ?

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Perception and Scene Understanding

Sense and interpret the environment to locate an apple and estimate its properties.

Sensor Data Acquisition: for example, RGB-D camera (color + depth), LiDAR, stereo camera, tactile and proximity sensors...

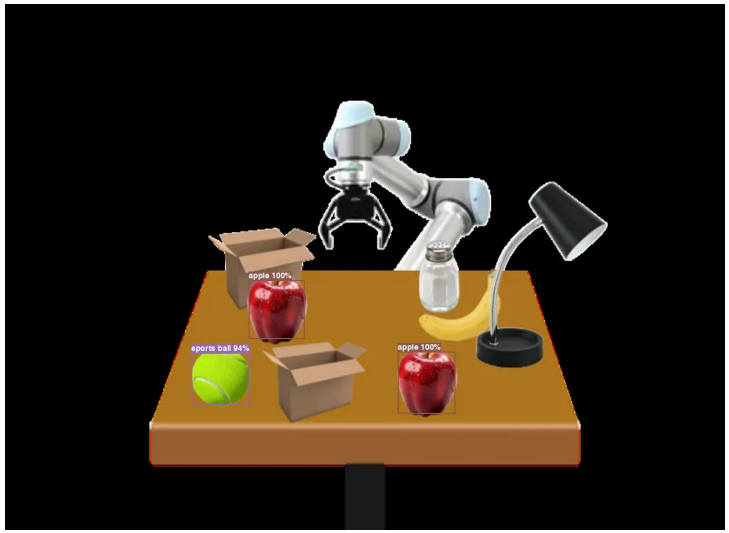


`https://huggingface.co/spaces/nielsr/dpt-depth-estimation`

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Perception and Scene Understanding

detect and classify objects in the image

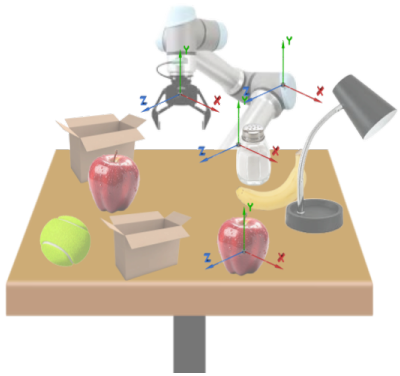


YOLO v3: `https://v-iashin.github.io/detector`

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Perception and Scene Understanding

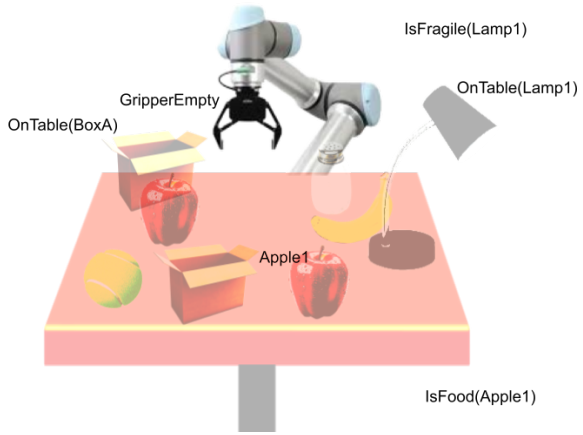
- **3D Localization:** Using the depth map, the robot computes the apple's 3D position in its camera coordinate frame.
- This point is transformed into world coordinates using the robot's calibration data.
- Uncertainty of estimation influences planning (e.g., grasp robustness).



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Task Interpretation - Task Planning

Translate the scene into a symbolic representation



The representation (e.g., using PDDL or STRIPS domain) is designed to be used by the high-level symbolic planner that finds a sequence of actions or policy to achieve the goal.

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Task Interpretation - Task Planning

- A planner translate the goal condition (“pick up an apple”) into a concrete sequence of actions.
- Possible representation:

Goal: (holding(apple) = true)

State Features: apple_on(table), gripper_empty ...

Actions: move_to(apple), grasp(apple)...

Is this a sufficient representation for our setting ? What if we have more than one way to achieve the goal (here, more than one apple)?



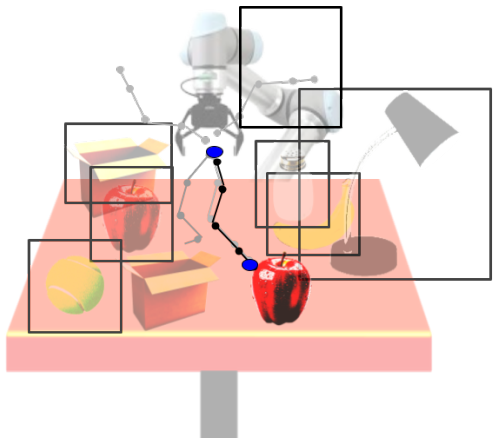
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Task Interpretation - Motion Planning

A motion planner is used to compute a collision-free trajectory for the robot's arm to reach the apple.

The planner ensures:

- desired position is reached (the gripper's approach vector aligns with the apple)
- no collision occurs with the table or nearby objects.
- kinematic feasibility (joint limits, reachability, etc.) is satisfied.

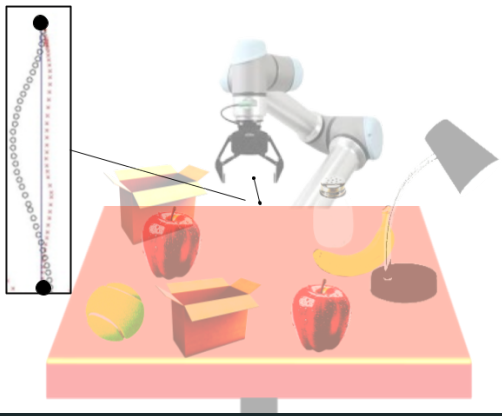


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Control and Execution

”Move the arm, close the gripper, and lift the apple”

- **Trajectory Execution** - low-level controller (e.g., PID, MPC) uses feedback from joint encoders and force sensors to follow the planned trajectory.
- **Force control loop** - tactile sensors measure contact pressure so that the controller can regulate the grasp force.
- **Vision-based feedback** - (visual servoing) refines the alignment if necessary; the apple disappears from the table view and appears in gripper view.



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Uncertainty Management and Error Handling

- If the apple is occluded, the robot performs active exploration (moves head/camera).
- If the robot's gripper misses the apple, it may trigger a failure recovery behavior (e.g., retry with a new pose estimate).
- If grasp fails (slip detected or object dropped), the system may replan or retry.



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Learning and Adaptation

The objective is to improve performance over time.

- **Reinforcement Learning:** The robot can learn grasp strategies through trial and error based on success rates.
- **Imitation Learning:** The robot can learn from demonstrations of humans picking apples.
- **Active Perception:** If uncertainty in pose estimation is high, the robot may reposition its camera to obtain a better view before acting.





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Human–Robot Interaction

- If the robot works with a human:
- It may confirm the object by asking: “Should I pick up this apple?”
- Safety layers ensure compliant motion when humans are nearby (collision avoidance, dynamic replanning).



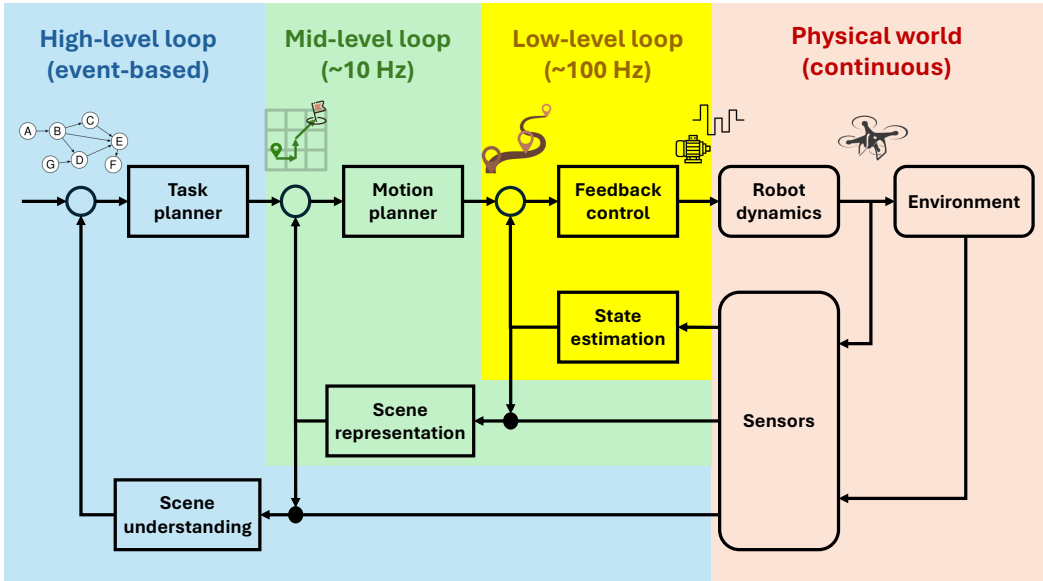
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Designing Complex systems

Perception–planning–control pipeline

Stage	Subsystem	Representative Methods
Perception	Vision & tactile sensing	CNNs, point-cloud segmentation, SLAM
Task Planning	Symbolic reasoning	PDDL, STRIPS, LLM-based planners
Motion Planning	Continuous optimization	RRT*, CHOMP, TrajOpt
Control	Dynamics and feedback	PID, impedance, MPC
Learning	Adaptive behavior	RL, imitation learning, active perception
HRI	Natural interaction	Speech, gesture, safety-aware motion





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Studying Robotics

Why Study Robotics ?



Figure 3: Yesterday's robots

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Why Study Robotics ?



Figure 4: Way in the future robots

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Why Study Robotics ?

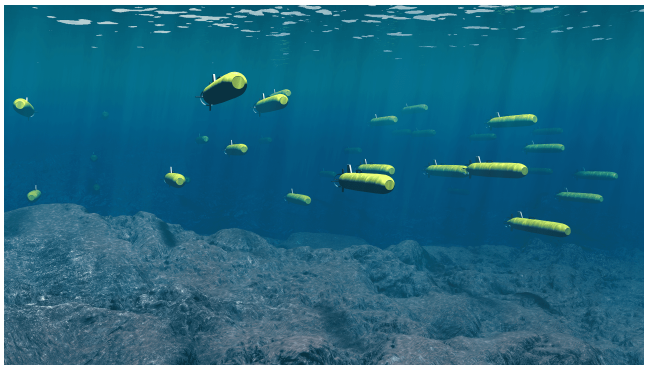


Figure 5: Today's robots

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Why Study Robotics ?





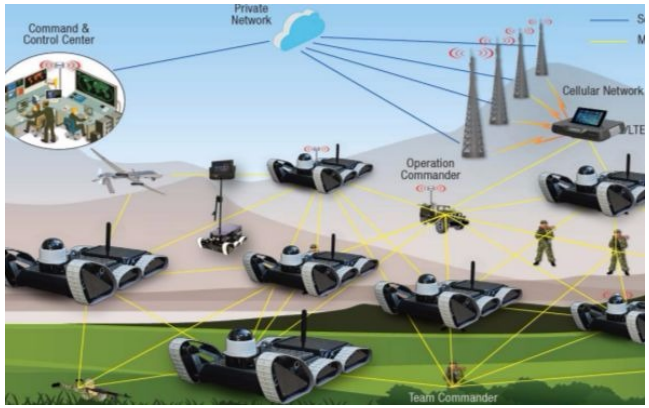
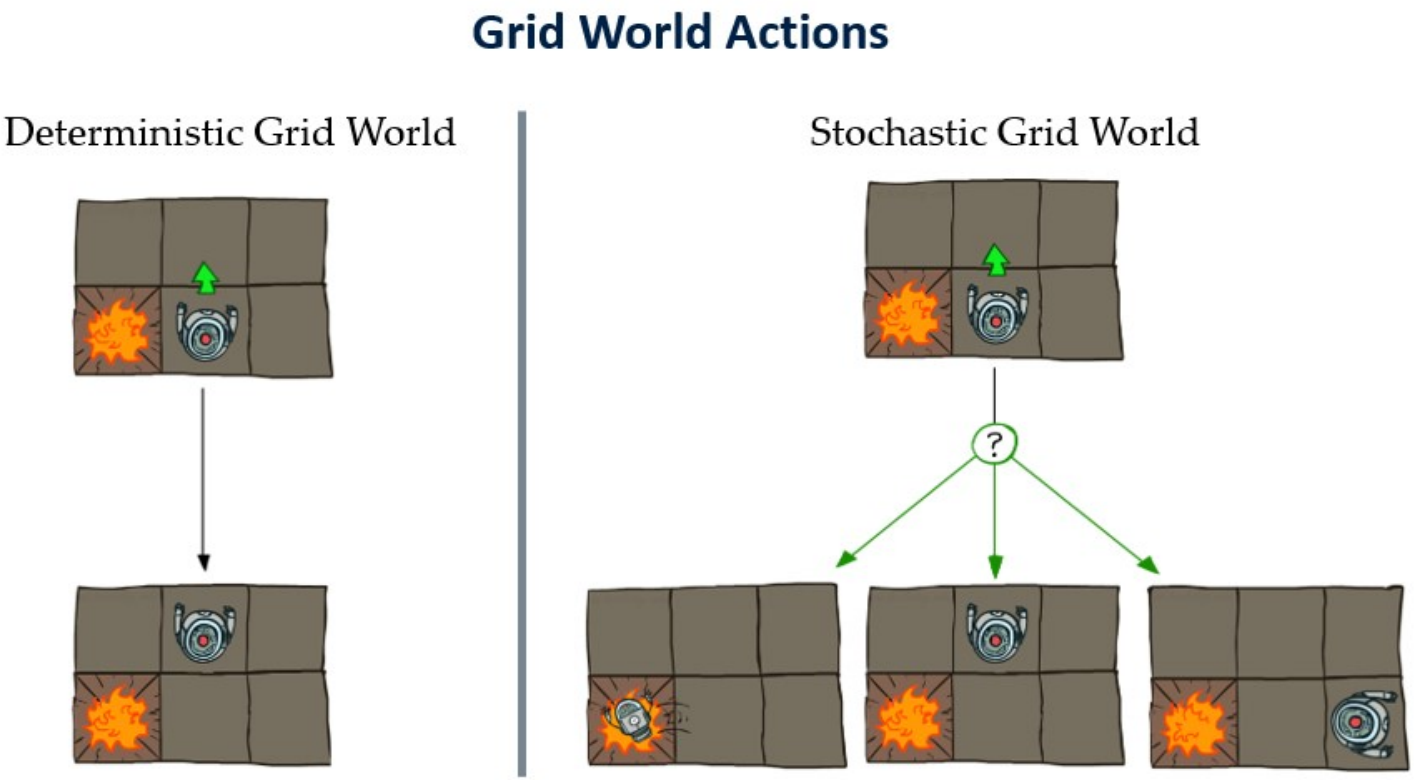


Figure 6: Today's robots

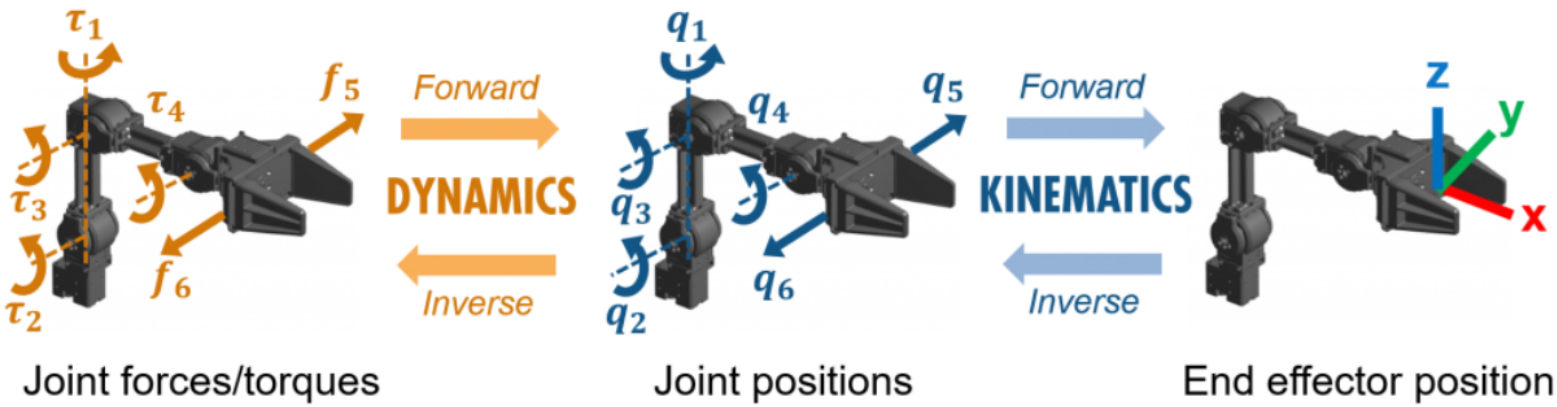
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Why take this course if you already took IAI?



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Why take this course if you already took a robotics course?

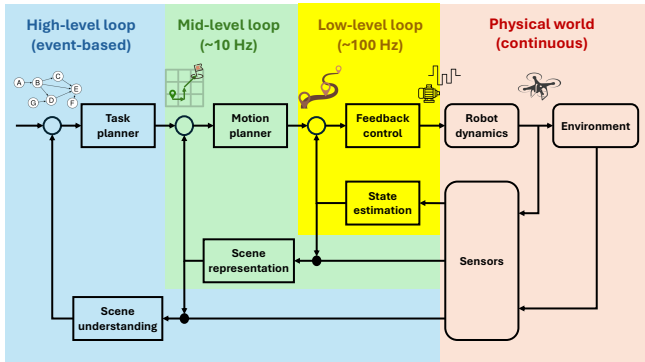


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Why take this course even if you are not really interested in robotics?

- It's not just about machines that move - it's about understanding how intelligent systems perceive, reason, and act in the real world.
- This requires algorithms to deal with uncertainty, noise, and delayed feedback.

Our focus is on designing **intelligent and adaptive [embodied] systems** that can make decisions and act effectively in complex environments.



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Why take this course if foundation models can be used to solve robotics ?



NVIDIA's humanoid based on foundation models

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Key Point:

Robots are quickly moving out of the 'factory' and into the 'real-world'. Soon there will be robots for the masses, but we aren't there yet...

Autonomy: Complex decision making in a complex world
Sensing, Thinking, Acting

A robot does not have to be a single physical object : Any combination of sensors and actuators can be thought of as a robot (smart home, factory, ...)

Autonomy



Challenges:

- How to model and perceive the world?
- How to process information and execute actions?
- How to reason and plan in the face of uncertainty?

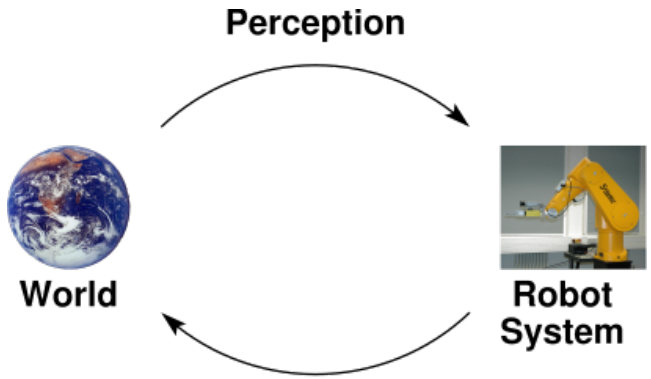
The field of robotics provides methods that address each challenge and architectures that combine them.

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Autonomy

What does it mean to be autonomous ?

- **Action (Actuation)**
  - Locomotion: wheels (e.g., differential drives)
  - Manipulation: arms, grippers, etc.
- **Perception (Sensors)**
  - Proprioception (internal: IMU (inertial measurement unit), encoders , etc,)
  - Exteroception (external: cameras, scanners)
- **Cognition (Control)**
  - From reactive to proactive
  - From finite state machines to cognitive robotics



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Bluey and Bingo



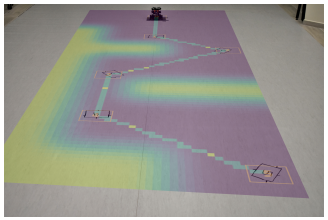
https://youtu.be/n4rh2jD8OkY

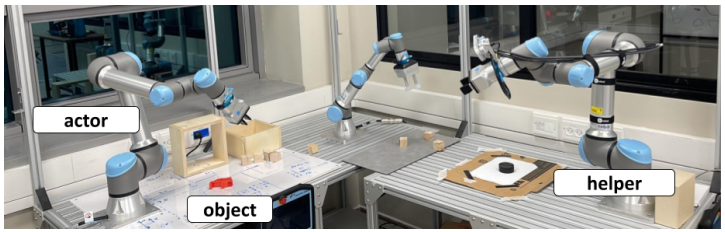
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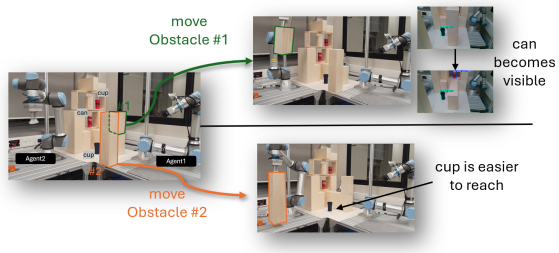
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Work done at the CLAIR lab











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What is this class about ?

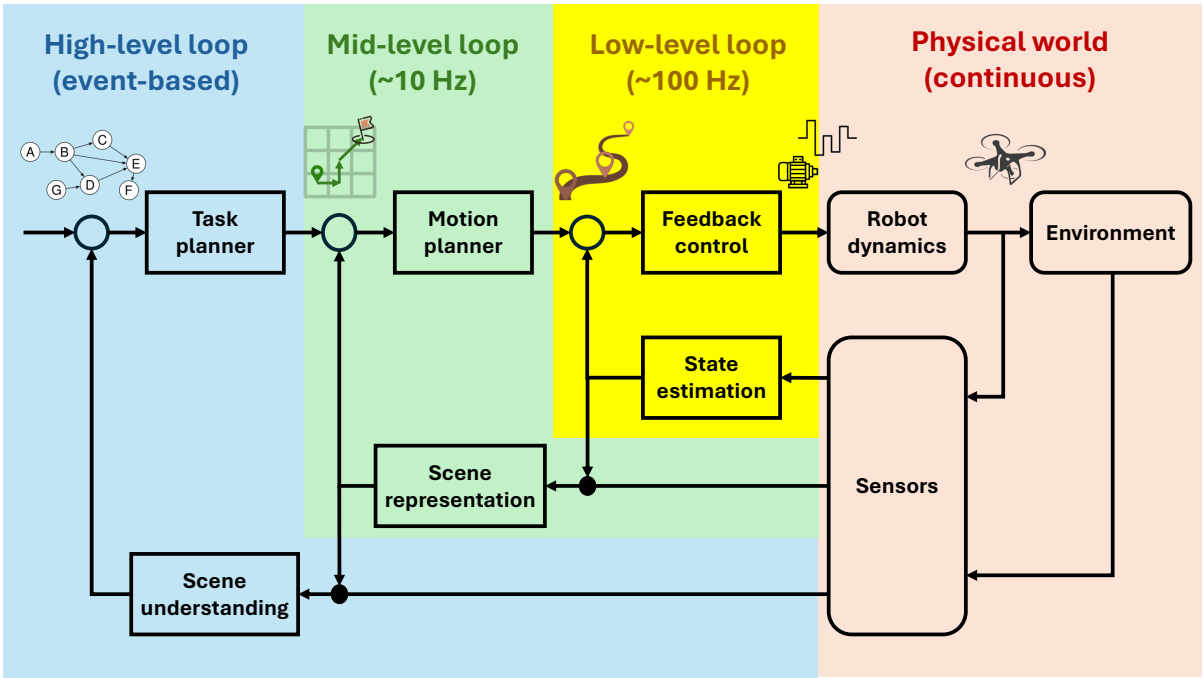
- From observations to decisions
- From low-level to high-level reasoning
- From single agent to multi-Agent
- From theory to practice to theory ...



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Our Focus

Integrating task planning, motion planning, and control.



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Course plan and relationship to IAI

Week	Lecture Topic	IAI
1	Introduction Approaches to decision making in AI	Intro Reactive vs. cognitive
2	Components of a robotic agent and Three-Layered Decision Making in Robotics	NA NA
3	Task planning for robotics and Reactive control	Search
4	Motion planning	Search
5	Planning under partial observability and noisy sensing	MDP, POMDP, MCTS, RL
6	Planning under partial observability and noisy sensing	—
7	Integrated task and motion planning	—
8	Integrated task and motion planning (contd.)	—
9	ML and RL in robotics	ML, RL
10	Integrated learning and planning for robotic agents	—
11	Integrated learning and planning (contd.)	—
12	Multi-robot systems	Multi-agent Search
13	Conclusion	NA

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Staff and Schedule

- **Teaching:** Sarah Keren
  `sarahk@cs.technion.ac.il`
- **TA:** Guy Azran
  `guy.azran@campus.technion.ac.il`
- **Schedule:**
  - Lesson: Wednesday: 14:30-16:30
  - Lab / tutorial: Monday 16:30-17:30
  - Reception Hours: by request.
- **Communication with staff:**
  - Most communication should be done by the representative.

> The subject of emails sent to us should begin with **02360006 AIR 2025-26** followed by your topic of discussion. It will otherwise be discarded.

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Course Structure

- Weekly 2-hour class and a 1-hour tutorial.
- We will have 3 homework assignments and an exam.
- Quiz (MAGEN) week 8

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Useful Links

- 3rd Summer School on Cognitive Robotics (2019): https://sites.usc.edu/cognitive-robotics/schedule/
- TASP - Technion Autonomous Systems Program https://www.youtube.com/channel/UCaOlOHpGQc5Rza5bXzgLObw/featured
- MIT Robotics seminar: https://www.youtube.com/channel/UCK2tKzmSFFnpFhUXtRKjvnQ?app=desktop

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Robotics at the Technion



Figure 7: Some of the Technion's robotics researchers

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Summary

- We motivated the study of robotics
- We highlighted the relationship between task planning, motion planning and control

What next ?

- Examine different AI Approaches and their relevance to robotics



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Extras

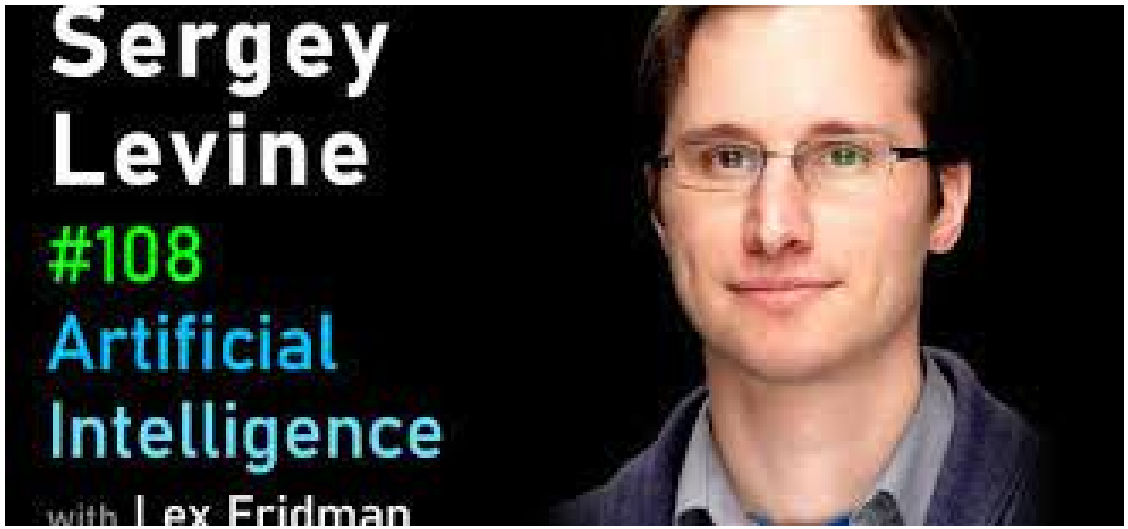
Single Robot Autonomy - Are we there yet ?





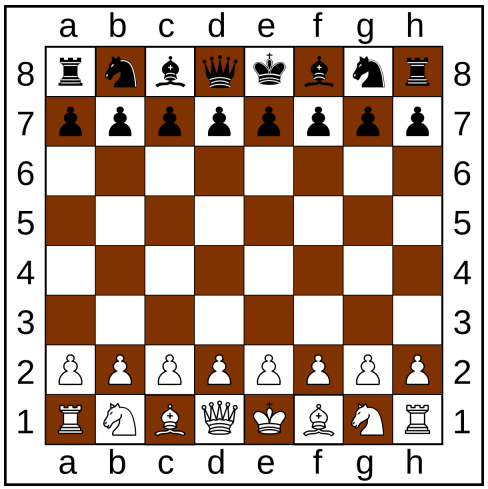
https://www.youtube.com/live/AtNGJeXTXu4?si=90REz2DGfoH4F54Q

Single Robot Autonomy - Are we there yet ?



https://youtu.be/kxi-_TT_-Nc?t=164

Single Robot Autonomy - Are we there yet ?





https://youtu.be/QaCuEv_7lfs?t=19739

George Konidaris: https://cs.brown.edu/~gdk/



1994 AAAI Mobile Robot Competition and Exhibition



Figure 1. The Participating Robots in the Competition.

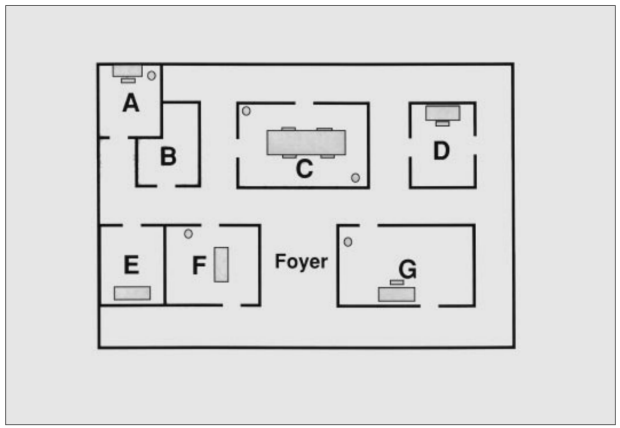


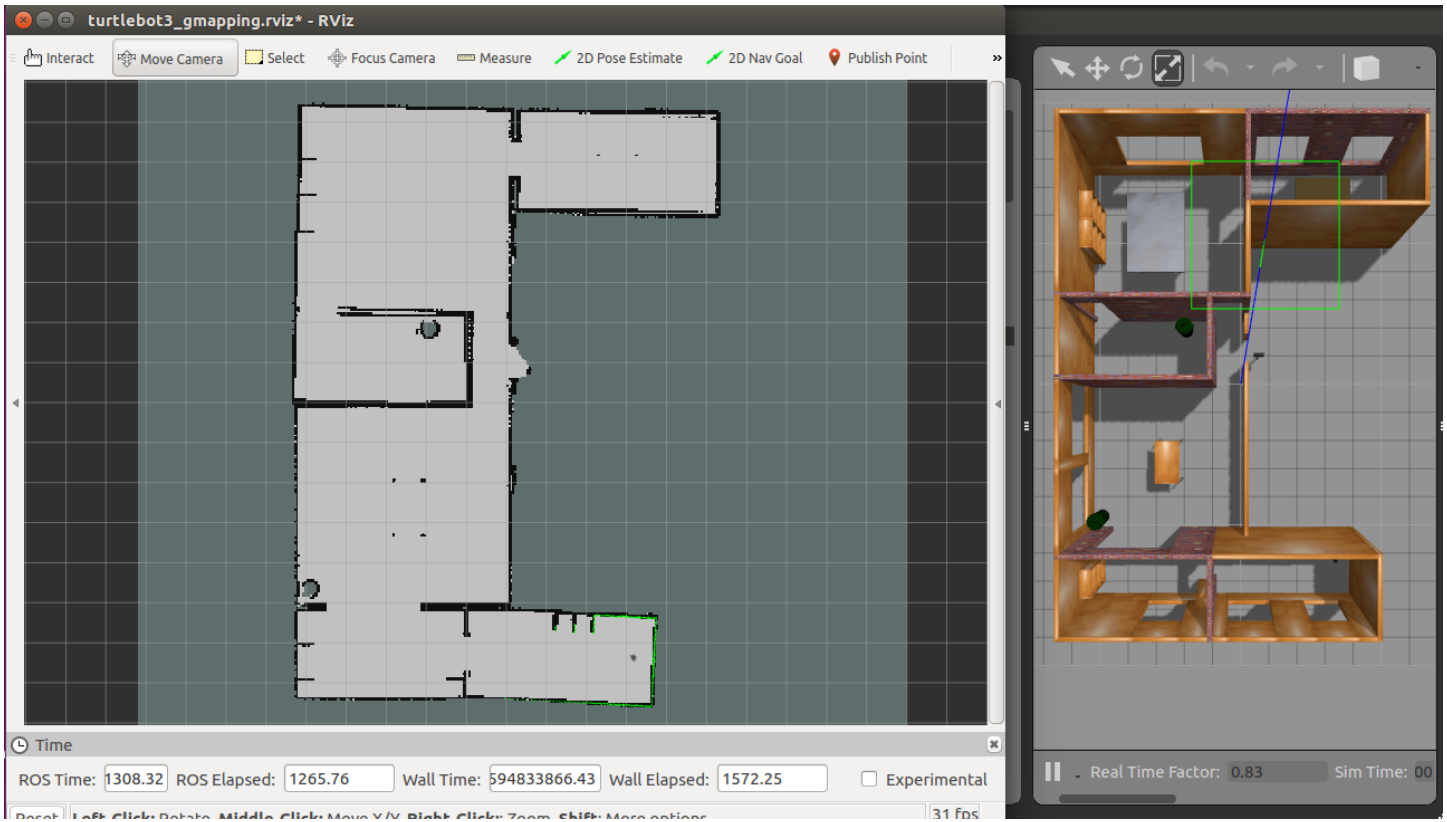
Figure 2. The Competition Arena.

- **Office Delivery**:
  navigate from one room to another within a large arena of offices, corridors, and foyers.
- **Office Cleanup**: the same arena was littered with cups, empty soda cans, and paper wads.
  - robots had to recognize and collect as many objects as possible in the allotted time and deposit them in near by trash bins.
  - several trials, with the main difference being the density of trash placed in the offices

https://doi.org/10.1609/aimag.v23i1.1603//
https://ojs.aaai.org//index.php/aimagazine/article/view/1130

Turtlebot3 SLAM

Extras



Search and Rescue



Manipulation

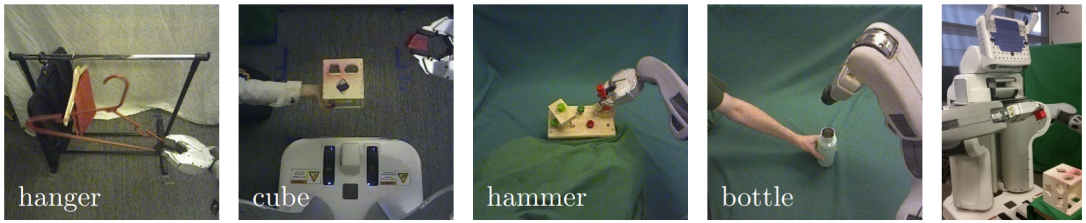
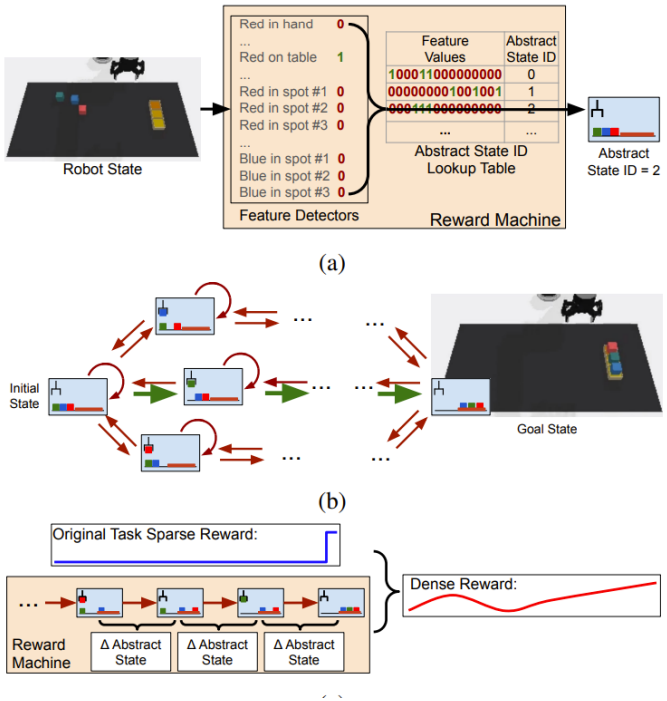
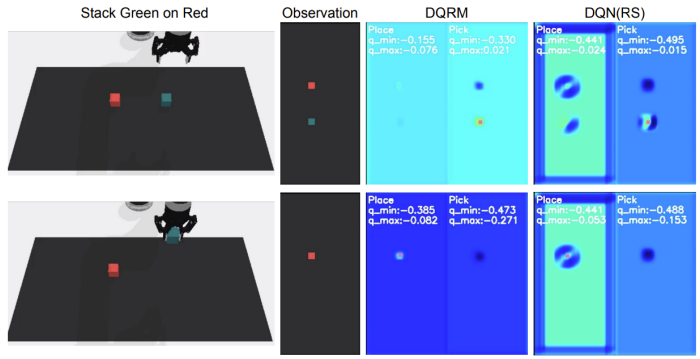


Figure 1: Our method learns visuomotor policies that directly use camera image observations (left) to set motor torques on a PR2 robot (right).

End-to-End Training of Deep Visuomotor Policies
Sergey Levine, Chelsea Finn, Trevor Darrell, Pieter Abbeel(2016)
`https://jmlr.org/papers/volume17/15-522/15-522.pdf`

Manipulation





Reward Machines for Vision-Based Robotic Manipulation. Camacho et al. (2021)

`https://ieeexplore.ieee.org/abstract/document/9561927`

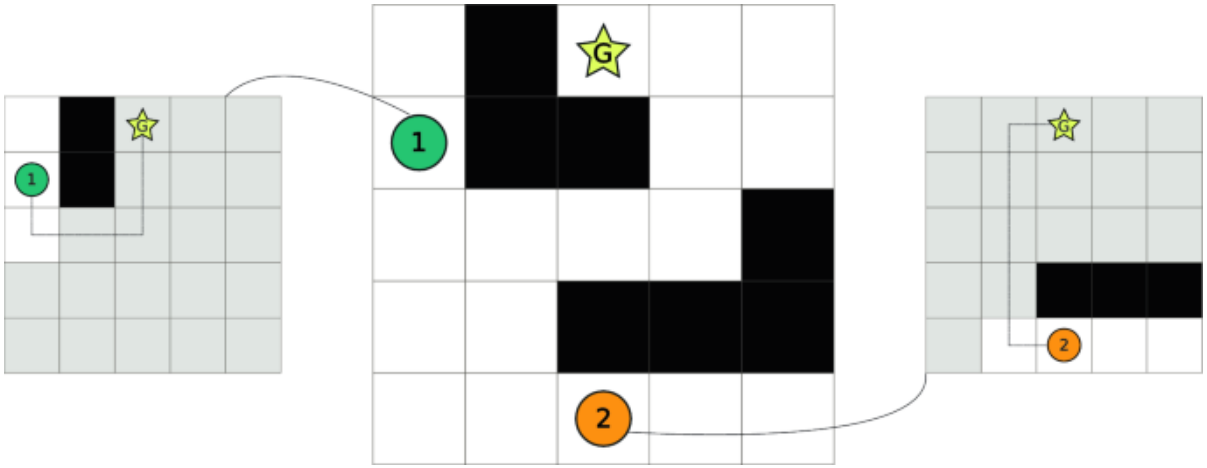
Cooking



https://www.youtube.com/watch?v=JNOk1rylDpU
https://arxiv.org/abs/1802.08705
https://www.youtube.com/watch?v=5R-xL9YmdR0

Multi-Robot

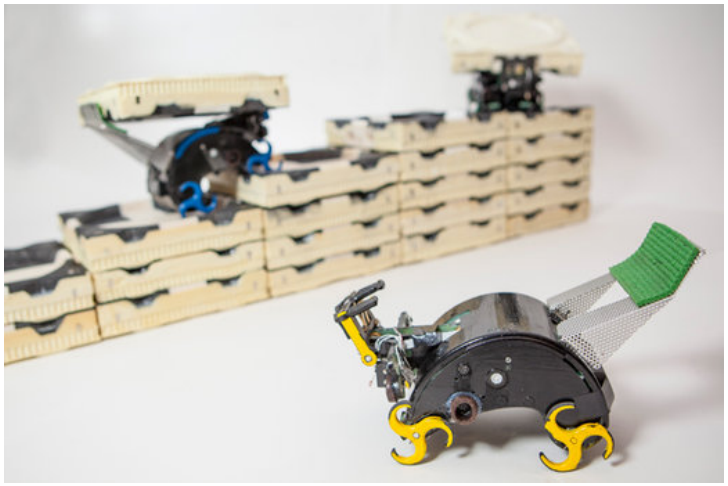




Optimizing Multi-Robot Communication under Bandwidth Constraints - Marcotte et al. (2019) `https://april.eecs.umich.edu/media/pdfs/marcotte2019auro.pdf`

Multi-Robot





AI and Robotics: Safety

South Korea: Man crushed to death by robot that mistook him for a box The man was reportedly inspecting the mechanical arm when the accident happened.

Thursday 9 November 2023 06:09, UK



https://news.sky.com/story/south-korea-man-crushed-to-death-by-robot-that-mistook-him-for-a-box-13003635